

Computer-Assisted Existence and Fixed-Wall Radial Stability of a Dirichlet-Confined Massive Skymion

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Abstract

We give a computer-assisted proof for a massive $SU(2)$ Skyrme hedgehog on a fixed de Sitter static background with a prescribed spherical Dirichlet boundary. Unlike prior rigorous work on the flat, whole-space Skymion, the validated problem is massive, positively curved, and confined at finite radius. At dimensionless parameters $\mu^2 = 1$, $\lambda = 1/400$, and $x_w = 4$, an exact-rational Newton–Kantorovich certificate proves existence and local uniqueness in an augmented ball of radius $1/250$. The certified solution is strictly decreasing, its wall slope lies in $[-0.09465, -0.08746]$, and its dimensionless rotor inertia lies in $[21.149, 48.921]$. Invertibility of the certified linearization also gives a locally unique real-analytic branch on an open neighborhood of the mass, curvature, and wall-radius parameters, although no numerical width is claimed. A second authenticated certificate proves a fixed-wall radial frequency gap $\widehat{\omega}_{\text{rad}} \geq 1/5$. Finally, exact endpoint jets and interval bounds on six third-derivative norms give the sharp inertia-transform asymptotic $\mathcal{N}^{(k)}(p) = O(p^{-3})$ for $k = 0, 1, 2$, with nonzero oscillatory leading coefficient fixed by the boundary second jet. This tail is boundary-induced and is not presented as an intrinsic ultraviolet prediction of an unsupported Skymion. We claim neither global uniqueness nor a self-gravitating, nonspherically stable, dynamically supported, or smoothly confined solution.

1 Introduction

Skymions provide a canonical setting in which topology, nonlinear field equations, collective coordinates, and curved geometry meet [1, 2]. Rigorous analysis is not new here. McLeod and Troy proved existence and uniqueness for the spherically symmetric flat-space Skyrme problem [3]. Creek, Donninger, Schlag, and Snelson later gave a rational-arithmetic, mildly computer-assisted existence proof and proved co-rotational linear stability on the whole space [4]. The present work is therefore not the first computer-assisted existence or stability proof for a Skymion.

The nearest model variants are also established. Static and gravitating Skymions in de Sitter space, spinning collective-coordinate corrections, and solitons confined by a mirror or cavity have been investigated numerically [5, 6, 7]. Nguyen and Nguyen explicitly survey finite-radius boundary conditions for the original model [8]. These sources show that positive curvature, finite radius, and validated numerics are not individually novel. The narrower gap addressed here is their conjunction for one massive, fixed-de-Sitter, unit-charge Dirichlet hedgehog, including an augmented finite-boundary proof and certified derived observables.

This paper validates one deliberately narrow boundary-value problem. We fix the de Sitter metric rather than solving the Einstein equations, impose a spherical Dirichlet wall strictly inside the cosmological horizon, and choose one dimensionless parameter triple. The central result is local: an exact-rational Newton certificate proves a unique solution inside a stated neighborhood of a rational spline. The certificate also controls enough of the solution to prove strict monotonicity, a negative wall slope, and positive finite collective inertia. Invertibility of the certified derivative then gives a qualitative real-analytic solution branch

Work	Model	Rigorous result or distinction
McLeod–Troy [3]	Flat, whole-space spherical Skymion	Existence and uniqueness.
Creek et al. [4]	Flat, whole-space co-rotational Skymion	Rational computer-assisted existence and linear stability.
Nguyen–Nguyen [8]	Finite-radius boundary survey	Numerical/global boundary analysis, not an interval certificate for the present massive curved problem.
This work	Massive fixed-de-Sitter hedgehog on a finite Dirichlet ball	Local uniqueness in an augmented graph-norm ball, fixed-boundary radial gap, analytic local parameter branch, and authenticated boundary-induced inertia-transform tail.

Table 1: Closest rigorous and finite-radius comparisons.

on an open neighborhood of the three parameters; the paper does not assign that neighborhood a numerical width.

Three features make the validation more than a certified shooting zero. First, the origin is regular singular in the usual radial variables. We replace it by a parameter-dependent Volterra chart and validate the regular slope family and its first two sensitivities. Second, the unknown shooting slope is included in an augmented boundary map. A coercive Dirichlet Jacobi operator and a nonzero scalar Schur complement provide its inverse, while row-local Green estimates and a center-equation Hessian identity close the nonlinear tube. Third, two consequences are authenticated against the same profile certificate: a radial fluctuation gap and the exact large-frequency tail of a finite optical transform of the rotor-inertia density.

The spectral result also makes the modeling boundary visible. The optical inertia weight has a quadratic zero at the Dirichlet wall, so boundary-aware integration by parts gives an oscillatory p^{-3} tail with nonzero coefficient. General endpoint expansions for finite Fourier and Hankel transforms are classical [9, 10]; our contribution is not that regularity controls decay. It is the certified endpoint order, coefficient, and global derivative envelope for the exact nonlinear solution.

Computer-assisted contraction and radii-polynomial proofs for nonlinear boundary-value problems are also established methodology [11, 12]. Relative to Creek et al., the technical changes are the finite regular-origin-to-Dirichlet matching problem, the unknown slope as an augmented variable, a massive curved coefficient set, and authenticated wall, inertia, radial-gap, and inertia-tail outputs. The proof-relevant implementation is a substantial exact-rational code base, inventoried in section 5; it is not described as a small checker. All numerical values used in theorem statements are outward enclosures derived from archived rational data; converged floating-point plots are used only for orientation.

The paper is organized as follows. Section 2 defines the fixed background, hedgehog problem, and central theorem. Section 3 develops the computer-assisted proof. Section 4 proves the radial gap and spectral tail. Section 5 identifies the source-pinned validation surface, archives, and reproduction path. Section 6 states the mathematical role of the Dirichlet boundary and the limits of the result.

2 Model, boundary-value problem, and certified solution

We work on a prescribed four-dimensional de Sitter static patch, with metric signature $(+ - - -)$,

$$ds^2 = N(r) dt^2 - N(r)^{-1} dr^2 - r^2 d\Omega_2^2, \quad N(r) = 1 - \frac{r^2}{R_{\text{dS}}^2}.$$

The matter field is $U: \mathcal{M} \rightarrow SU(2)$, with $R_\mu = U^\dagger \nabla_\mu U$, and our massive Skyrme convention is

$$S_{\text{Sk}} = \int_{\mathcal{M}} d^4x \sqrt{-g} \left[-\frac{f_\pi^2}{16} \text{Tr}(R_\mu R^\mu) + \frac{1}{32e^2} \text{Tr}([R_\mu, R_\nu][R^\mu, R^\nu]) + \frac{f_\pi^2 m_\pi^2}{16} \text{Tr}(U + U^\dagger - 2) \right].$$

In particular, m_π is the mass of the linearized pion field; the coefficient of the last term is not the convention in which this mass is $\sqrt{2}m_\pi$.

For the static hedgehog

$$U(r, \hat{x}) = \cos F(r) + i \boldsymbol{\tau} \cdot \hat{x} \sin F(r),$$

introduce

$$x = e f_\pi r, \quad \mu = \frac{m_\pi}{e f_\pi}, \quad \lambda = \frac{1}{(e f_\pi R_{\text{dS}})^2}, \quad N(x) = 1 - \lambda x^2,$$

and abbreviate $s = \sin F$ and $u = x^2 + 8s^2$. The bulk static mass is

$$M_{\text{bulk}} = \frac{4\pi f_\pi}{e} \int_0^{x_w} \mathcal{E}(x) dx,$$

where

$$\mathcal{E} = \frac{1}{8} N u (F')^2 + \frac{1}{4} \sin^2 F + \frac{\sin^4 F}{2x^2} + \frac{\mu^2 x^2}{4} (1 - \cos F).$$

Here and below a prime denotes d/dx . Its Euler–Lagrange equation is

$$(NuF')' = \left(4N(F')^2 + 1 + \frac{4 \sin^2 F}{x^2} \right) \sin(2F) + \mu^2 x^2 \sin F. \quad (1)$$

The apparent singularity at $x = 0$ is removable on the regular hedgehog branch described below.

The prescribed Dirichlet boundary

We truncate the static domain at $x = x_w$, strictly inside the Killing horizon $x_c = \lambda^{-1/2}$, and impose the Dirichlet constraint $U|_{x_w} = \mathbf{1}_2$. In the unit-charge hedgehog sector the boundary data are therefore

$$F(0) = \pi, \quad F(x_w) = 0. \quad (2)$$

No wall equation of motion is imposed in this paper: (2) is prescribed data, not a claim that a particular finite-tension membrane dynamically realizes the boundary.

For the certified problem,

$$\mu^2 = 1, \quad \lambda = \frac{1}{400}, \quad x_w = 4, \quad x_c = 20. \quad (3)$$

Thus the complete boundary-value problem is (1)–(3) on $[0, 4]$. This parameter point is a validation benchmark, not a phenomenological fit. The choice $\mu = 1$ keeps the pion and Skyrme length scales comparable, while $\sqrt{\lambda}x_w = 1/5$ gives the rational wall lapse $N(x_w) = 24/25$ and keeps the complete domain well inside the horizon. It therefore tests nonzero mass, curvature, and finite confinement without a near-horizon singular limit. The paper does not infer a distinct de Sitter effect from this single point. A qualitative local continuation family is proved in corollary 2.2, without an explicit box width.

Regularity class and the augmented chart

We record the function spaces because they are part of the scope of the uniqueness statement. Set

$$a = \frac{1}{16}, \quad I = [a, 4].$$

The removable origin is treated in the variables

$$t = x^2, \quad \pi - F = xu(t), \quad p_b(t) = \frac{d(\pi - F)}{dx}.$$

For a positive shooting slope $b = -F'(0)$, the validated Volterra chart has

$$p_b(t) = b - 3c(b)t + t^2\rho_b(t), \quad \|\rho_b\|_{L^\infty(0,a^2)} \leq 20. \quad (4)$$

The cubic coefficient appearing here is exactly

$$c(b) = \frac{b \left[\mu^2 - 4\lambda + \left(\frac{4}{3} - 24\lambda \right) b^2 + \frac{8}{3}b^4 \right]}{10(1 + 8b^2)}. \quad (5)$$

Consequently

$$|F(x) - (\pi - bx + c(b)x^3)| \leq 4x^5, \quad |F'(x) - (-b + 3c(b)x^2)| \leq 20x^4. \quad (6)$$

These bounds hold on $[0, a]$. The Volterra contraction selects one regular solution inside this weighted ball for each certified b ; it does not assert uniqueness among all conceivable singular-origin function classes.

Let F_b^{org} denote that regular origin solution, and define its cutoff data

$$\Phi(b) = F_b^{\text{org}}(a), \quad \Gamma(b) = (F_b^{\text{org}})'(a).$$

On I , the augmented formulation consists of (1) together with

$$F(a) = \Phi(b), \quad F(4) = 0, \quad F'(a) = \Gamma(b). \quad (7)$$

On the positive-radius interval the Dirichlet correction space is

$$X_D = \{h \in W^{2,\infty}(I) : h(a) = h(4) = 0\}.$$

The exact center slope used by the certificate is

$$\bar{b} = \frac{1462763601523}{925827031517}.$$

Let $\widehat{F}$ be the archived globally C^2 , piecewise-rational polynomial profile, and set

$$\begin{aligned} \chi_a(x) &= \frac{4-x}{4-a}, & \chi_4(x) &= \frac{x-a}{4-a}, \\ \bar{u} &= \widehat{F} - \chi_a \widehat{F}(a) - \chi_4 \widehat{F}(4), & \bar{F} &= \bar{u} + \chi_a \Phi(\bar{b}). \end{aligned} \quad (8)$$

Thus $\bar{F}(a) = \Phi(\bar{b})$ and $\bar{F}(4) = 0$ exactly; neither endpoint residual of the generated spline is discarded. The inverse used by the certificate is built instead from the Jacobi operator at the raw rational spline $\widehat{F}$. Put

$$p(x, F) = N(x)(x^2 + 8 \sin^2 F), \quad W(x, F) = \sin^2 F + \frac{2 \sin^4 F}{x^2} + \mu^2 x^2 (1 - \cos F),$$

and let

$$A_0 h = -(P_0 h')' + Q_0 h, \quad P_0 = p(x, \widehat{F}), \quad Q_0 = W_{FF}(x, \widehat{F}) + \frac{1}{2} p_{FF}(x, \widehat{F})(\widehat{F}')^2 - (p_F(x, \widehat{F})\widehat{F}')'. \quad (9)$$

The certificate proves that the graph norm

$$\|h\|_{A_0} = \|A_0 h\|_{L^\infty(I)}$$

is equivalent to the usual $W^{2,\infty}$ norm on X_D , and hence controls h and h' pointwise. The endpoint-corrected $\overline{F}$ remains the center of the nonlinear chart below. The certificate separately encloses the coefficient change from $D\mathcal{G}(\widehat{F}) = A_0$ to $D\mathcal{G}(\overline{F})$; this is the principal contribution to the reported derivative-mismatch bound Z_0 .

For completeness, the local neighborhood used by the theorem can be stated without suppressing the shooting variable. Let H be the certified homogeneous Jacobi solution

$$A_0 H = 0, \quad H(a) = \Phi'(\overline{b}), \quad H(4) = 0.$$

In local coordinates $(w, \beta) \in X_D \times \mathbb{R}$, the physical profile and shooting slope are

$$F_{w,\beta} = \overline{F} + w + \beta H + \chi_a [\Phi(\overline{b} + \beta) - \Phi(\overline{b}) - \Phi'(\overline{b})\beta], \quad b = \overline{b} + \beta. \quad (10)$$

This chart enforces the two value conditions in (7) for every coordinate pair. The derivative condition remains the scalar augmented equation; at a zero it gives the C^1 join to the regular origin family. With $\omega = 3/4$, the certified closed ball is

$$\mathcal{B}_r = \{(w, \beta) : \max(\|w\|_{A_0}, \omega|\beta|) \leq r\}, \quad r = \frac{1}{250}. \quad (11)$$

In particular, its slope interval is exactly

$$\overline{b} + \left[-\frac{2}{375}, \frac{2}{375}\right] = \left[\frac{546684696508091}{347185136818875}, \frac{550388004634159}{347185136818875}\right]. \quad (12)$$

Theorem 2.1 (Certified massive Dirichlet profile). *For the fixed-background boundary-value problem (1)–(3), there exists a pair $(w_*, \beta_*) \in \mathcal{B}_{1/250}$ such that $F_* = F_{w_*, \beta_*}$ solves (1) and all three conditions in (7). Adjoining the certified regular-origin piece gives a profile on $[0, 4]$ obeying (2). This pair is the only solution represented in $\mathcal{B}_{1/250}$.*

The exact-rational Newton certificate, using $\omega = 3/4$, twelve-term Taylor–Lagrange trigonometric enclosures, and outward rounding on the grid with denominator 10^{18} , bounds the Newton defect Y , the approximate-inverse derivative mismatch Z_0 , and the Schur-composed nonlinear Hessian bound Z_2 by

$$Y \leq \frac{287}{125000}, \quad Z_0 \leq \frac{39}{500000}, \quad Z_2 \leq \frac{78733}{1000}. \quad (13)$$

For $r = 1/250$, the corresponding radii polynomial and contraction bound satisfy

$$p(r) = Y + (Z_0 - 1)r + \frac{1}{2}Z_2 r^2 < -\frac{107}{100000}, \quad q(r) = Z_0 + Z_2 r < \frac{31501}{100000} < 1. \quad (14)$$

Therefore the self-map and contraction inequalities both close.

Moreover, F_* is strictly decreasing on $[0, 4]$. Its wall slope lies in the exact archived rational enclosure

$$F'_*(4) \in \left[-\frac{94644972770042989}{10^{18}}, -\frac{43733242881735721}{5 \cdot 10^{17}} \right] \subset \left[-\frac{1893}{20000}, -\frac{4373}{50000} \right]. \quad (15)$$

In particular, the wall slope is nonzero and negative. Finally, define the dimensionless interior rigid-rotor inertia by

$$\mathcal{I}[F] = \frac{2\pi}{3} \int_0^4 x^2 \sin^2 F \left[\frac{1}{N} + 4(F')^2 + \frac{4 \sin^2 F}{Nx^2} \right] dx. \quad (16)$$

Directed interval quadrature on the same Newton tube, together with the regular-origin estimate, proves the exact conservative enclosure

$$\frac{21149}{1000} \leq \mathcal{I}[F_*] \leq \frac{48921}{1000}. \quad (17)$$

Thus the matter contribution to the physical collective inertia, $I_{\text{bulk}} = \mathcal{I}[F_*]/(e^3 f_\pi)$, is finite and strictly positive.

The paper uses the source-consistent combined AU.1/AU.2 archive `experiments/skymion_au2_global_tail_exact_certificate.json`. It contains the profile, auxiliary, Green-factor, representer, and exact AU.1 output records used by this theorem. Its SHA-256 digest is

1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9.

Equations (13)–(17) are conservative readable consequences of that archive; the self-map and contraction decisions themselves are made before decimal formatting. A historical AU.1-only extraction is not a release dependency because its embedded source snapshot predates the current audit driver.

Corollary 2.2 (Local analytic parameter branch). *There is an open neighborhood*

$$U \subset \{(m^2, \ell, R) : m^2 > 0, \ell > 0, \ell R^2 < 1\}$$

of $(1, 1/400, 4)$ and a real-analytic family of regular solutions $F_{m^2, \ell, R}$ of the corresponding Dirichlet problem on $[0, R]$. After pullback to a fixed interval, this family is locally unique near F_* . On a possibly smaller neighborhood, every member is strictly decreasing, has nonzero negative derivative at the wall, and has finite positive rotor inertia.

Proof. Set $\theta = (m^2, \ell, R)$ and $\theta_0 = (1, 1/400, 4)$. Pull $[a, R]$ back to $I = [a, 4]$ by

$$s_\theta = \frac{R - a}{4 - a}, \quad x_\theta(\xi) = a + s_\theta(\xi - a).$$

Let $\mathcal{Y} = L^\infty(I) \times \mathbb{R}$ and use the base graph coordinates $X = (w, \beta) \in \mathcal{X}$ from (10). Define the parameter-dependent endpoint lift on the fixed interval by

$$\tilde{F}_{X, \theta} = \bar{F} + w + \beta H + \chi_a \left[\Phi_\theta(\bar{b} + \beta) - \Phi_{\theta_0}(\bar{b}) - \Phi'_{\theta_0}(\bar{b})\beta \right]. \quad (18)$$

Because $H(a) = \Phi'_{\theta_0}(\bar{b})$ and all three correction directions vanish at $\xi = 4$, this gives $\tilde{F}_{X, \theta}(a) = \Phi_\theta(\bar{b} + \beta)$ and $\tilde{F}_{X, \theta}(4) = 0$ exactly. At $\theta = \theta_0$ it reduces to (10).

For clarity, write the pulled-back coefficients as

$$\begin{aligned} \tilde{p}_\theta(\xi, F) &= (1 - \ell x_\theta(\xi)^2)(x_\theta(\xi)^2 + 8 \sin^2 F), \\ \tilde{W}_\theta(\xi, F) &= \sin^2 F + \frac{2 \sin^4 F}{x_\theta(\xi)^2} + m^2 x_\theta(\xi)^2 (1 - \cos F). \end{aligned}$$

The transformed interior operator is therefore

$$\begin{aligned}\mathcal{G}_\theta^{\text{pb}}(\tilde{F}) = & -s_\theta^{-2}\partial_\xi\left(\tilde{\mathfrak{p}}_\theta(\xi, \tilde{F})\partial_\xi\tilde{F}\right) \\ & + \frac{1}{2}s_\theta^{-2}\partial_F\tilde{\mathfrak{p}}_\theta(\xi, \tilde{F})(\partial_\xi\tilde{F})^2 + \partial_F\tilde{W}_\theta(\xi, \tilde{F}).\end{aligned}\tag{19}$$

Thus the interior residual and the physical derivative-matching condition define

$$\mathcal{R} : \mathcal{O}_X \times \Theta \longrightarrow \mathcal{Y}, \quad \mathcal{R}(X, \theta) = \left(\mathcal{G}_\theta^{\text{pb}}(\tilde{F}_{X, \theta}), s_\theta^{-1}\partial_\xi\tilde{F}_{X, \theta}(a) - \Gamma_\theta(\bar{b} + \beta)\right). \tag{20}$$

The positive-radius coefficients make (20) real analytic in (X, θ) while $\ell R^2 < 1$. The regular-origin Volterra map is jointly analytic in its shooting slope and θ and remains a strict contraction after shrinking Θ . Its fixed point is therefore analytic, so the cutoff maps Φ_θ and Γ_θ are jointly analytic as used in (20).

At the exact zero X_* , the certificate gives

$$\|I - BD_X\mathcal{R}(X_*, \theta_0)\| < 0.31501 < 1.$$

The reference block inverse B is an isomorphism, so the Neumann series makes $D_X\mathcal{R}(X_*, \theta_0)$ invertible. The analytic implicit-function theorem now gives the asserted local branch and local uniqueness. The certificate bounds F'_* strictly below zero on every origin and positive-radius cell, including the wall. These inequalities persist under sufficiently small C^1 perturbations. Positivity and finiteness of the inertia then persist by continuity of its integrand on the compact pulled-back interval. $\square$

Claim boundary. Theorem 2.1 is an existence and local-uniqueness theorem for one prescribed parameter point and one explicitly certified augmented neighborhood. Corollary 2.2 proves existence on an open parameter neighborhood but does not give its numerical width or one uniform Newton ball. Neither statement is a global uniqueness theorem. At the certified point the geometry is fixed: the result does not solve the Einstein–Skyrme equations or bound gravitational backreaction. The wall is an imposed ideal Dirichlet boundary, and the theorem does not provide a support action, membrane dynamics, junction conditions, finite-stiffness response, or wall inertia. Nor does static existence establish linear or nonlinear dynamical stability, an off-center solution, an open-system approximation, or a de Sitter observer-algebra statement.

3 Computer-assisted validation

This section gives the analytic skeleton of the computer-assisted proof. The proof has two logically separate parts. On the regular-origin patch we use a uniform Volterra contraction, with the initial slope as a parameter. On the positive-radius interval we use an augmented Newton–Kantorovich argument for the profile and that slope together. Candidate splines may be generated numerically, but every premise used below is recomputed with rational interval arithmetic. In particular, convergence of a floating-point shooting method is not a premise of the theorem.

Throughout this section

$$a = \frac{1}{16}, \quad c = 4, \quad I = [a, c], \quad N(x) = 1 - \frac{x^2}{400}.$$

For convenient reference, write the profile equation in variational form as

$$\begin{aligned}\mathcal{G}(F) &= -(\mathfrak{p}(x, F)F')' + \frac{1}{2}\mathfrak{p}_F(x, F)(F')^2 + W_F(x, F), \\ \mathfrak{p}(x, F) &= N(x)(x^2 + 8\sin^2 F), \\ W(x, F) &= \sin^2 F + \frac{2\sin^4 F}{x^2} + x^2(1 - \cos F).\end{aligned}\tag{21}$$

The apparent singularity at $x = 0$ is handled before this formula is used on I .

3.1 The regular-origin chart

Set

$$g(x) = \pi - F(x) = xu(t), \quad t = x^2, \quad p(t) = g'(x) = u(t) + 2tu'(t).$$

Thus $p(0) = u(0) = b$, where $b = -F'(0) > 0$ is the shooting parameter, and

$$u_p(t) = \int_0^1 p(ts^2) \, ds.\tag{22}$$

Introduce the entire even kernels

$$\text{sinc}_1(z) = \frac{\sin z}{z}, \quad \text{sinc}_2(z) = \frac{\sin(2z)}{2z},$$

with their values at zero defined by continuity. Direct substitution in the Euler–Lagrange equation gives a nonsingular Volterra map. One useful form is

$$\begin{aligned}A_p(t) &= u_p(t) \text{sinc}_1(\sqrt{t} u_p(t)), \\ D_p(t) &= 1 + 8A_p(t)^2, \\ Q_p(t) &= 2u_p(t) \text{sinc}_2(\sqrt{t} u_p(t)) [1 + 4A_p(t)^2 + 4N(\sqrt{t})p(t)^2] - tA_p(t), \\ (\mathcal{V}p)(t) &= \frac{1}{2tN(\sqrt{t})D_p(t)} \int_0^t Q_p(s) \, ds,\end{aligned}\tag{23}$$

where the last expression is continued analytically to $t = 0$. Formula (23) is especially useful for validation: all trigonometric dependence occurs through entire power series, and integration divides the coefficient of t^j by the exact integer $2(j + 1)$.

Lemma 3.1 (Uniform regular-origin family). *There is a common rational slope interval containing the certified center $\bar{b} = 1462763601523/925827031517$ on which (23) has a unique fixed point in the weighted balls*

$$p_b(t) = b - 3\gamma(b)t + t^2 r_b(t), \quad \|r_b\|_\infty \leq R, \quad 0 \leq t \leq a^2.\tag{24}$$

Here $\gamma(b)$ is the cubic coefficient obtained from the exact constant and linear coefficient identities. The fixed point is twice continuously differentiable in b . Consequently

$$\Phi(b) = F_b(a), \quad \Gamma(b) = F'_b(a)$$

and their first two derivatives are rigorously enclosed. The same validation is repeated on a wider slope interval containing the complete Newton slope tube used below.

Proof. The checker translates (23) by the first two exact Taylor coefficients and evaluates it on the weighted remainder ball in (24). It verifies three strict inequalities: the Volterra denominator stays positive, the image maps the ball into itself, and the Lipschitz constant is smaller than one. The parameterized contraction theorem then gives the fixed-point family. First- and second-order interval automatic differentiation is applied to the same contraction equation, not to finite differences of endpoint boxes. At the narrow center family the verified data are

$$b \in [1.579953, 1.579954], \quad R = \frac{13}{10}, \quad a = \frac{1}{16}.$$

For the nonlinear tube the checker uses the larger remainder ball and slope interval recorded by that tube and rechecks the same three inequalities. The exact rational coefficient identities, power-series remainder bounds, and the interval tables for both slope ranges are delegated to the origin certificate artifact. This lemma asserts uniqueness only in the displayed weighted fibers; it does not assert unrestricted uniqueness of a singular initial-value problem. $\square$

For the center profile, a separately validated quintic origin expansion is nested strictly inside the cubic fiber in (24). Uniqueness in that fiber identifies the two fixed points. This sharper point expansion is used to seed the rational positive-radius candidate, while the cubic family supplies the b -derivative bounds required by the augmented proof.

3.2 Rational center and augmented boundary map

The candidate on I is a globally C^2 , piecewise-quintic spline $\widehat{F}$ on a 43-cell graded rational mesh. Every node, coefficient, and join is a rational number. Numerical integration is used only to propose these data. The checker verifies the joins and reevaluates all residuals from the archived coefficients.

Let

$$\chi_a(x) = \frac{c-x}{c-a}, \quad \chi_c(x) = \frac{x-a}{c-a}.$$

Rather than assume that the rational candidate has exact boundary values, set

$$\bar{u} = \widehat{F} - \chi_a \widehat{F}(a) - \chi_c \widehat{F}(c), \quad \bar{F} = \bar{u} + \chi_a \Phi(\bar{b}). \quad (25)$$

Then $\bar{F}(a) = \Phi(\bar{b})$ and $\bar{F}(c) = 0$ exactly. The interval width in $\Phi(\bar{b})$ is retained in every subsequent coefficient and residual calculation.

We regard the endpoint values as part of the chart and define

$$\mathcal{N}(F, b) = (\mathcal{G}(F), F'(a) - \Gamma(b)), \quad F(a) = \Phi(b), \quad F(c) = 0. \quad (26)$$

The verified nonlinear residual is evaluated in a cancellation-preserving form: after multiplication by x^2 , it is collected into exact rational polynomial amplitudes multiplying

$$1, \quad \cos F, \quad \sin(2F), \quad \cos(2F), \quad \cos(4F).$$

Each full expression is summed as a centered Taylor model before its range is taken. This step is essential near the origin, where separately ranging the divergence terms loses several orders of magnitude.

Lemma 3.2 (Authenticated center residual). *The endpoint-corrected center (25) and the scalar boundary residual in (26) admit exact rational interval enclosures on all 43 cells. The spline joins, endpoint lifts, five-harmonic identity, and all Taylor remainders are checked independently of the candidate generator.*

Proof. This is a finite exact-arithmetic verification once the harmonic identity has been obtained by symbolic differentiation. The proof archive stores the rational profile coefficients and the accepted aggregate outputs. On replay, the checker reconstructs every cellwise residual from those coefficients and recomputes the scalar bound entering the final Newton defect; it does not accept a decimal summary or a serialized cellwise table as proof input. $\square$

3.3 Dirichlet inverse and scalar Schur complement

The reference inverse is built from the self-adjoint Dirichlet operator at the raw rational spline,

$$A_0 h = -(P_0 h')' + Q_0 h, \quad h(a) = h(c) = 0, \quad (27)$$

where $P_0 = p(x, \widehat{F})$ and Q_0 is obtained by differentiating (21). The fixed positive comparison function

$$v_*(x) = \frac{8}{(x - 33/16)^2 + 4}$$

satisfies a certified Barta inequality $A_0 v_*/v_* \geq \alpha > 0$. The ground-state identity therefore yields

$$\langle h, A_0 h \rangle = \int_I P_0 v_*^2 [(h/v_*)']^2 dx + \int_I \frac{A_0 v_*}{v_*} h^2 dx, \quad (28)$$

so A_0 is injective and obeys the maximum principle.

For sharp pointwise control, the checker does not use the coarse energy bound alone. Exact rational splines $\widehat{u}$ and $\widehat{v}$, vanishing at a and c , respectively, define the factored Green parametrix

$$\widehat{G}(x, s) = \frac{\widehat{u}(\min\{x, s\})\widehat{v}(\max\{x, s\})}{w_0}, \quad (29)$$

where w_0 is the certified oriented principal Wronskian. The residuals $A_0 \widehat{u}$ and $A_0 \widehat{v}$ control both the Wronskian drift and the operator defect.

Lemma 3.3 (Green resolvent bounds). *The operator A_0 has a bounded inverse from $L^\infty(I)$ to $W^{2,\infty}(I) \cap W_0^{1,\infty}(I)$. The parametrix in (29) has certified defect $\varepsilon < 0.089015$, and the exact inverse obeys*

$$\|A_0^{-1}\|_{L^\infty \rightarrow C^0} < 0.228138, \quad \|A_0^{-1}\|_{L^\infty \rightarrow C^1} < 1.43213, \quad \|A_0^{-1}\|_{L^\infty \rightarrow C^2} < 59.3126. \quad (30)$$

The same construction yields a separate row bound for each source cell.

Proof. The checker verifies $P_0 > 0$ and the Barta inequality on an adaptively refined rational partition; the archived exact lower bound satisfies $\alpha > 1.02359$. Applying A_0 to the integral operator defined by (29) gives $I + E$ with $\|E\| < 1$. Hence

$$A_0^{-1} = \widehat{T}(I + E)^{-1},$$

and a Neumann correction converts the exact row integrals of $\widehat{G}$, $\partial_x \widehat{G}$, and the second-derivative jump term into (30). The audit replay reconstructs the Barta leaves, Wronskian bounds, fundamental-solution residuals, and row integrals from the archived rational splines. The archive records their accepted aggregate bounds and source digests rather than serializing the transient adaptive tables. $\square$

It remains to invert the one-dimensional shooting block. Define

$$\tau(f) = (A_0^{-1} f)'(a).$$

If κ solves $A_0\kappa = 0$, $\kappa(a) = 1$, $\kappa(c) = 0$, Green's identity gives the exact representation

$$\tau(f) = \frac{1}{P_0(a)} \int_I \kappa(x) f(x) \, dx. \quad (31)$$

An independently validated rational approximation to κ supplies the trace bound; it is not inferred from the Schur solution it will later test.

Let H solve

$$A_0H = 0, \quad H(a) = \Phi'(\bar{b}), \quad H(c) = 0, \quad S = H'(a) - \Gamma'(\bar{b}). \quad (32)$$

Use local coordinates

$$F = \bar{F} + w + \beta H + \chi_a [\Phi(\bar{b} + \beta) - \Phi(\bar{b}) - \Phi'(\bar{b})\beta], \quad (33)$$

with $w(a) = w(c) = 0$. Both value boundary conditions are then exact for every (w, β) .

Lemma 3.4 (Augmented inverse). *The scalar Schur factor in (32) is nonzero; the final audit gives $S > 2.8918$. For the reference block built from A_0 , the inverse has the form*

$$w = A_0^{-1}r, \quad \beta = \frac{s - \tau(r)}{S}, \quad (34)$$

for an interior residual r and derivative-boundary residual s .

Proof. The candidate H is assembled from a rational regular-sensitivity spline and a rational fundamental solution so that its right endpoint vanishes exactly. The checker subtracts the image of its full residual using the independently certified trace representer (31). The corrected interval remains strictly positive. Since lemma 3.3 makes A_0 invertible, elementary block elimination gives (34). The replay recomputes the residual, L^1 representer, trace correction, and Schur enclosures before accepting the aggregate output. Keeping the representer calculation separate is important: using H itself to certify the norm of τ would be circular. $\square$

We equip the Dirichlet component with the graph norm

$$\|w\|_{A_0} = \|A_0w\|_\infty$$

and the augmented coordinates with

$$\|(w, \beta)\|_{\mathcal{X}} = \max\{\|w\|_{A_0}, \omega|\beta|\}, \quad \omega = \frac{3}{4}. \quad (35)$$

The C^j estimates in (30), together with the corresponding corrected bounds for H , make this graph ball a pointwise C^1 tube. This is the bridge needed for the sign claims below; an energy ball alone would not control F' pointwise.

3.4 Reduced Hessian and the radii polynomial

A direct interval bound for $D^2\mathcal{G}$ is valid but too coarse because it charges the global C^2 inverse norm in every cell. The closing estimate uses the equation satisfied by each first variation. If

$$A_0h = g_h, \quad A_0k = g_k,$$

put

$$\begin{aligned} m &= \mathfrak{p}_{FF}F' + \mathfrak{p}_{xF}, \\ z &= W_{FFF} - \mathfrak{p}_{FF}F'' - \mathfrak{p}_{xFF}F' - \frac{1}{2}\mathfrak{p}_{FFF}(F')^2. \end{aligned}$$

In the following identity, P_0, Q_0 , and P'_0 are the coefficients of the fixed reference operator at $\widehat{F}$, whereas $\mathfrak{p}_F, m, z$ are evaluated at the profile F in the Newton tube.

Lemma 3.5 (Center-equation Hessian identity). *For twice differentiable first variations h, k ,*

$$\begin{aligned} D^2\mathcal{G}(F)[h, k] &= \frac{\mathfrak{p}_F}{P_0}(kg_h + hg_k) - \mathfrak{p}_F h'k' + \left(\frac{\mathfrak{p}_F P'_0}{P_0} - m\right)(hk' + kh') \\ &\quad + \left(z - 2\frac{\mathfrak{p}_F Q_0}{P_0}\right)hk. \end{aligned} \tag{36}$$

Consequently the nonlinear Hessian can be bounded cell by cell using only the local C^0 and C^1 Green rows and the known sources g_h, g_k .

Proof. Expand (21) twice. In the resulting symmetric expression replace both second derivatives using

$$h'' = \frac{Q_0 h - P'_0 h' - g_h}{P_0}, \quad k'' = \frac{Q_0 k - P'_0 k' - g_k}{P_0}.$$

Collecting terms gives (36); the factor 2 in the final coefficient comes from the two substitutions. This is an exact algebraic identity, not an integration-by-parts estimate. The checker bounds both the direct and reduced formulas and takes the smaller certified bound on each cell. The symbolic expansion is human-checkable; the checker recomputes the coefficient ranges and selected cellwise bounds during the audit replay. $\square$

The scalar component also retains cell localization. If $K_J = \int_J |\widehat{\kappa}| \, dx$ on a source cell J and $\epsilon_\kappa = C_0 \|A_0 \widehat{\kappa}\|_\infty$, then nonnegative cell bounds b_J obey

$$T(b) = \frac{1}{P_0(a)} \sum_J (K_J + \epsilon_\kappa |J|) b_J. \tag{37}$$

Thus a large near-origin Hessian value is not multiplied by a global trace constant over the whole domain.

Let B denote the exact bounded inverse of the reference augmented block defined by A_0, τ , and S . Its norm bounds are certified through the rational Green and Schur parametrices above. Using the norm (35), the checker establishes

$$\begin{aligned} Y &\geq \|B\mathcal{N}(\bar{F}, \bar{b})\|_X, \\ Z_0 &\geq \|I - B\mathcal{D}\mathcal{N}(\bar{F}, \bar{b})\|_{X \rightarrow X}, \\ \|B[\mathcal{D}\mathcal{N}(X) - \mathcal{D}\mathcal{N}(\bar{X})]\|_{X \rightarrow X} &\leq Z_2 \|X - \bar{X}\|_X \quad \text{for } \|X - \bar{X}\|_X \leq r. \end{aligned} \tag{38}$$

The Z_0 calculation includes the coefficient mismatch between the A_0 -based reference inverse and the derivative at the endpoint-corrected center. The affine origin lift in (33), including the certified bounds on Φ'' and Γ'' , is included in Z_2 .

Lemma 3.6 (Newton–Kantorovich closure). *For $\omega = 3/4$ and $r = 1/250$, exact rational comparison gives the conservative bounds*

$$Y \leq 0.002296, \quad Z_0 \leq 0.000078, \quad Z_2 \leq 78.733. \quad (39)$$

The radii polynomial and contraction factor satisfy

$$\rho(r) = Y + (Z_0 - 1)r + \frac{1}{2}Z_2r^2 < -0.00107, \quad q(r) = Z_0 + Z_2r < 0.31501. \quad (40)$$

Hence (26) has a unique zero in the closed X -ball of radius $1/250$ about the certified center.

Proof. Equations (38) imply that

$$\mathcal{T}(X) = X - BN(X)$$

maps the closed radius- r ball into itself when $\rho(r) < 0$, and has Lipschitz constant at most $q(r)$. The strict inequalities in (40) therefore give a contraction. The numbers in (39) are outward decimal summaries of exact fractions; the checker compares the fractions themselves. On replay it reconstructs the 43-cell Hessian bounds, local Green rows, weighted-trace contributions, affine-lift bounds, and operator-mismatch bounds. The exact certificate stores the accepted fractions and all rational spline inputs needed for that replay, but not the transient adaptive tables. $\square$

The uniqueness in lemma 3.6 is local and coordinate-explicit: it is uniqueness in the augmented graph ball, not global uniqueness among all finite-energy hedgehog profiles.

3.5 Post-Newton physical checks

The contraction theorem is complete before any physical sign is asserted. The checker then propagates the same Newton ball to pointwise jets. On the origin patch, the Volterra enclosure gives $p(t) > 0$, hence $F'(x) < 0$. On each positive-radius cell, the local Green bounds, the corrected auxiliary, and the quadratic origin lift produce a tube enclosure for F' . Every such enclosure has a strictly negative upper endpoint. The final cell gives the certified wall interval

$$F'(4) \in [-0.09465, -0.08746]. \quad (41)$$

No sampled-grid interpolation is used in either sign test.

Finally, the dimensionless rotor inertia is the integral of the nonnegative density

$$j(x, F, F') = \frac{2\pi}{3} \left[\frac{x^2 \sin^2 F}{N(x)} + 4x^2 \sin^2 F (F')^2 + \frac{4 \sin^4 F}{N(x)} \right]. \quad (42)$$

The checker evaluates (42) by directed interval quadrature on the authenticated positive-radius cells. The origin Volterra family supplies a finite analytic upper bound on $[0, a]$; the lower bound there may be discarded because the density is nonnegative. A strictly positive lower sum on $[a, c]$ and the finite origin upper bound give

$$21.149 \leq \mathcal{I}_{\text{rot}} \leq 48.921. \quad (43)$$

Thus the validated solution is strictly decreasing, has a strictly negative wall slope, and has finite positive rotor inertia.

3.6 Exact arithmetic and proof provenance

The proof-relevant validation layer uses rational endpoints throughout. It evaluates π with a rational Machin formula, encloses sine and cosine with rational Taylor–Lagrange remainders, and carries interval automatic derivatives algebraically. For the final tube, selected nonnegative upper bounds are rounded upward to denominator 10^{18} , while interval lower and upper endpoints are rounded outward. Twelve-term trigonometric enclosures are used in the tube calculation. These rounding operations control denominator growth; they do not insert binary floating-point values into a theorem comparison.

The complete accepted input and output used by this paper are stored in

`experiments/skymion_a2_global_tail_exact_certificate.json`.

The archive contains the 43 rational cells for the profile, Schur auxiliary, left Green factor, and right Green/trace representer; exact output fractions; the command line; dependency versions; and SHA-256 digests of the checker sources. Its SHA-256 digest is

`1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9`.

The profile component also has a compact human-readable audit:

`experiments/skymion_newton_reduced_hessian_rounded_audit_result.json`.

The combined archive binds those outputs to the current source snapshot and the AU.2 endpoint calculation. The adaptive interval tables used in lemmas 3.1 to 3.6 are deterministically recomputed by the pinned checker. The archive retains the exact inputs and accepted aggregate outputs, and the body records the analytic identities and final outward enclosures needed to audit their role. Tamper-rejection tests separately perturb endpoints, splines, operator parameters, provenance links, and closing inequalities and require the checker to reject the altered input.

4 Certified radial and spectral consequences

Throughout this section, $F_\star := F_*$ denotes the exact profile in the certified Newton neighborhood, not the rational center spline, and $I_{\text{rot}} := \mathcal{I}[F_\star]$ is the dimensionless inertia defined in theorem 2.1. The two consequences below import the same authenticated wall-slope and inertia enclosures,

$$\begin{aligned} F'_\star(4) &\in [-0.094644972770043, -0.087466485763471], \\ I_{\text{rot}} &\in [21.149277305080, 48.920186135018]. \end{aligned} \tag{44}$$

They are therefore statements about the validated nonlinear solution. The combined profile-and-tail archive has SHA-256 digest

`1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9`.

Decimal endpoints in this section are outward-rounded displays of rational intervals; the archived fractions, rather than these displays, are used in the proof comparisons.

4.1 Fixed-wall spherical radial gap

Let $\tau = ef_\pi t$. Thus a mode with dimensionless frequency $\widehat{\omega}$ has Killing-time frequency $\omega_K = ef_\pi \widehat{\omega}$. Set

$$u(F, x) = x^2 + 8 \sin^2 F, \quad N(x) = 1 - \lambda x^2. \quad (45)$$

Linearization of the spherically symmetric Skyrme action about $F_\star$ gives the generalized Sturm–Liouville problem

$$\ell_\star \eta = \widehat{\omega}^2 \mathcal{W}_\star \eta, \quad \ell_\star = -\frac{d}{dx} \left(P_\star \frac{d}{dx} \right) + Q_\star, \quad (46)$$

where

$$P_\star = Nu(F_\star, x), \quad \mathcal{W}_\star = \frac{u(F_\star, x)}{N}. \quad (47)$$

Here $Q_\star$ is precisely the second-variation potential of the static functional used in the validated Jacobi problem. To specify the singular endpoint, let

$$\mathcal{H} = L^2((0, 4), \mathcal{W}_\star(x) dx) \quad (48)$$

and initially define

$$\mathfrak{q}_\star[\eta] = \int_0^4 \left(P_\star |\eta'|^2 + Q_\star |\eta|^2 \right) dx \quad (49)$$

on $C_c^\infty(0, 4)$. Its closure is the fixed-wall Friedrichs form. The associated self-adjoint operator in $\mathcal{H}$ is

$$\mathcal{A}_\star = \mathcal{W}_\star^{-1} \ell_\star, \quad \mathcal{A}_\star \eta = \widehat{\omega}^2 \eta. \quad (50)$$

The regular-singular classification below shows that eigenmodes select the physical branch $\eta(x) = O(x)$ at the origin; the closure enforces the Dirichlet trace at $x = 4$.

Proposition 4.1 (Authenticated fixed-wall radial stability). *For $\mu^2 = 1$, $\lambda = 1/400$, and $x_w = 4$, the exact profile satisfies*

$$\mathfrak{q}_\star[\eta] \geq \|\eta\|_{L^2(0,4)}^2 \quad (51)$$

on the closed Friedrichs form domain. The operator $\mathcal{A}_\star$ has compact resolvent in $\mathcal{H}$, and its lowest dimensionless spherical profile frequency obeys

$$\widehat{\omega}_{\text{rad}}^2 \geq \frac{1}{25}, \quad \widehat{\omega}_{\text{rad}} \geq \frac{1}{5}, \quad \omega_K \geq \frac{ef_\pi}{5}. \quad (52)$$

Proof sketch. Use the positive rational Barta witness

$$v(x) = \frac{8}{(x - 33/16)^2 + 4}. \quad (53)$$

On $[1/16, 4]$, adaptive interval evaluation through every exact-solution jet in the Newton tube gives

$$\inf_{[1/16, 4]} \frac{\ell_\star v}{v} > 1.0386099769. \quad (54)$$

The origin cannot be obtained by simply extending those boxes through the regular singular point. Instead, substituting $\pi - F_\star = xu(x^2)$ and using the exact profile equation removes all negative powers of x and gives

$$\inf_{[0, 1/16]} \frac{\ell_\star v}{v} > 36.8298881657. \quad (55)$$

The ground-state transform with v proves equation (51) first on $C_c^\infty(0, 4)$ and then, by closure, on the Friedrichs form domain. The regular-origin expansion gives, with $a_0 = 1 + 8b^2$,

$$P_\star = a_0 x^2 + O(x^4), \quad \mathcal{W}_\star = a_0 x^2 + O(x^4), \quad Q_\star = 2a_0 + O(x^2). \quad (56)$$

Thus the indicial roots of the eigenvalue equation are 1 and -2 . The x^{-2} branch is not in $\mathcal{H}$, so the origin is limit point and eigenmodes obey $\eta = O(x)$. At $x = 4$, where $P_\star > 0$, the Friedrichs closure gives the Dirichlet trace.

For completeness, the form embedding into $\mathcal{H}$ is compact. On every $[\delta, 4]$, the ground-state representation controls an H^1 norm and Rellich compactness applies. Near the origin, $\mathcal{W}_\star(x) \leq Cx^2$ and equation (51) give uniformly on form-bounded sets

$$\int_0^\delta \mathcal{W}_\star |\eta|^2 dx \leq C\delta^2 \mathfrak{q}_\star[\eta]. \quad (57)$$

The vanishing tail and the interior compactness prove compact resolvent.

Finally, $N(4) = 24/25$ and $0 \leq \sin^2 F_\star \leq 1$, whence

$$0 \leq \mathcal{W}_\star(x) \leq \frac{x_w^2 + 8}{N(x_w)} = 25. \quad (58)$$

The min–max principle for $\mathcal{A}_\star$, together with the generalized Rayleigh quotient, converts the unit static form gap into equation (52). The authenticated calculation used 109 positive-radius leaves, with maximum binary refinement depth five. $\square$

The exact radial-gap archive has SHA-256 digest

695310609d070f6dba2c678982ffc87472ed7c341813137fb90ad9dd9e1e6096.

It verifies the identity of the upstream spectral/profile archive before reconstructing the exact-solution tube. In particular, it authenticates the canonical spectral archive and sharp tube snapshot with respective digests

1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9
1781a2ff357f3b165d23e290eb403552d77d099a72b9e90920735e6a80b30431.

Remark 4.2 (Scope of the stability statement). Proposition 4.1 concerns only the $l = 0$ profile fluctuation with an ideal, nondynamical Dirichlet wall. It is not a theorem about a moving membrane, nonspherical or grand-spin sectors, the rotational collective coordinate, a holding apparatus, or metric perturbations. In particular, it is a radial spectral gap on the declared fixed-wall form domain, not full dynamical stability of a supported or self-gravitating Skyrminion.

4.2 Authenticated endpoint jets

Introduce the optical radius

$$x(y) = \frac{\tanh y}{\sqrt{\lambda}}, \quad Y = \operatorname{arctanh}(\sqrt{\lambda}x_w), \quad N_w = 1 - \lambda x_w^2. \quad (59)$$

Write the dimensionless inertia as

$$I_{\text{rot}} = \int_0^4 \rho_I(x) dx, \quad \rho_I(x) = \frac{2\pi}{3} x^2 \sin^2 F_\star \left[\frac{1}{N} + 4(F'_\star)^2 + \frac{4 \sin^2 F_\star}{Nx^2} \right]. \quad (60)$$

Define the two optical weights

$$\begin{aligned} W(y) &= \frac{\rho_I(x(y))}{x(y)^2} \frac{dx}{dy} \\ &= \frac{2\pi \sin^2 F_\star}{3\sqrt{\lambda}} \left[1 + 4\lambda \cosh^2 y F_{\star,y}^2 + 4\lambda \coth^2 y \sin^2 F_\star \right], \quad A(y) = W(y) \coth y. \end{aligned} \quad (61)$$

At the regular origin,

$$\begin{aligned} F_\star(x) &= \pi - bx + cx^3 + O(x^5), \\ c &= \frac{b \left[\mu^2 - 4\lambda + \left(\frac{4}{3} - 24\lambda\right)b^2 + \frac{8}{3}b^4 \right]}{10(1 + 8b^2)}, \end{aligned} \quad (62)$$

$$W(y) = w_0 y^2 + O(y^4), \quad A(y) = w_0 y + O(y^3), \quad w_0 = \frac{2\pi b^2(1 + 8b^2)}{3\lambda^{3/2}}. \quad (63)$$

At the wall, let $\sigma = |F'_\star(x_w)|$ and $\delta = Y - y$. Then

$$W(y) = w_Y \delta^2 + O(\delta^3), \quad A(y) = \coth(Y) w_Y \delta^2 + O(\delta^3), \quad (64)$$

$$w_Y = \frac{2\pi \sigma^2 N_w^2 (1 + 4N_w \sigma^2)}{3\lambda^{3/2}}, \quad W''(Y) = 2w_Y > 0. \quad (65)$$

The strict inequality follows from the authenticated negative wall slope in equation (44). Analytically, a nontrivial solution cannot have both $F_\star(x_w) = 0$ and $F'_\star(x_w) = 0$, since uniqueness for the regular wall initial-value problem would then give the vacuum solution.

For the present parameters,

$$Y = \operatorname{arctanh}(1/5) = 0.202732554054 \dots, \quad N_w = \frac{24}{25}, \quad (66)$$

and directed evaluation of the endpoint ledger gives

$$\begin{aligned} W''(Y) &\in [243.208686287, 286.156494827], \\ A''(Y) &\in [1216.04343143, 1430.78247414]. \end{aligned} \quad (67)$$

In particular, the leading wall amplitude is bounded away from zero.

4.3 Boundary-controlled inertia-transform tail

In terms of equation (61), define the finite optical transform of the inertia density

$$\mathcal{N}(p) = \frac{1}{p} \int_0^Y A(y) \sin(py) dy - \int_0^Y W(y) \cos(py) dy. \quad (68)$$

This is the profile-dependent numerator used in rotational matter-current form-factor constructions. We leave it unfiltered so that the result below isolates exactly what follows from the certified inertia distribution. This continuum expression, rather than a finite radial quadrature, is used at large frequency.

The relation between endpoint order and the asymptotics of finite Fourier or Hankel transforms is standard; see, for example, [9, 10]. We make no priority claim for that general mechanism. The result specific to this work is that the nonlinear profile certificate authenticates the quadratic wall zero, the nonzero coefficient in equation (67), and the global consequence below.

Proposition 4.3 (Sharp authenticated inertia-transform tail). *For the exact certified profile,*

$$\begin{aligned}\mathcal{N}(p) &= W''(Y) \frac{\sin(pY)}{p^3} + o(p^{-3}), \\ \mathcal{N}'(p) &= YW''(Y) \frac{\cos(pY)}{p^3} + o(p^{-3}), \\ \mathcal{N}''(p) &= -Y^2W''(Y) \frac{\sin(pY)}{p^3} + o(p^{-3}).\end{aligned}\tag{69}$$

Moreover, the certificate gives

$$(n_0, n_1, n_2) = (3.7575 \times 10^{10}, 3.7830 \times 10^{10}, 7.5660 \times 10^{10})\tag{70}$$

such that

$$|\mathcal{N}^{(k)}(p)| \leq n_k p^{-3}, \quad p \geq 1.\tag{71}$$

Proof sketch. For $m = 0, 1, 2$, set

$$M_m^W = \|\partial_y^3(y^m W)\|_{L^1(0,Y)}, \quad M_m^A = \|\partial_y^3(y^m A)\|_{L^1(0,Y)}.\tag{72}$$

The exact interval-AD certificate, with a separate regular-origin Volterra–Lie calculation, proves

$$\begin{aligned}(M_0^W, M_1^W, M_2^W) &\leq (2.5776861727020434 \times 10^8, 6.94657005661323 \times 10^6, 2.98269008103853 \times 10^5), \\ (M_0^A, M_1^A, M_2^A) &\leq (3.731716425863237 \times 10^{10}, 5.056441300063133 \times 10^8, 1.391944736624204 \times 10^7).\end{aligned}\tag{73}$$

These conservative values are upper displays of exact rational bounds.

Three integrations by parts are performed on the closed half interval $[0, Y]$. The endpoint jets in equations (63) and (64) cancel the lower-order boundary terms but retain the wall term proportional to $W''(Y)$. This point is essential: the zero extension of W has a jump in its second derivative at Y and must not be treated as a global $W^{3,1}$ function. The six norms in equation (73) bound the integral remainders and, after differentiating under the integral sign, give equation (71). Keeping the wall terms and applying the Riemann–Lebesgue lemma to the remainders gives the three sharp expansions in equation (69); derivatives retain the same power because they differentiate the endpoint phase pY .

The same explicit bookkeeping gives the three outward-rounded constants in equation (70). In the combined archive they are the upper endpoints of `numerator_derivative_coefficients`; no form-factor prefactor or frequency filter is applied to $\mathcal{N}$. $\square$

The endpoint and six-norm certificate has SHA-256 digest

1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9.

It binds the wall slope, inertia, endpoint ledger, all six derivative norms, and the continuum tail envelope to one certificate identity.

Remark 4.4 (Scope of the spectral statement). The p^{-3} coefficient is a property of this Dirichlet-confined inertia profile. A generic finite-pinning profile with nonzero wall value loses the quadratic cancellation and can have a slower tail; no smooth-confinement universality theorem is asserted. The transform (68) is defined directly from the certified inertia density. Identifying it with a detector bath factor would require a specified current, interaction, switching protocol, and open-system limit; none of those is asserted here.

5 Reproducibility and certificate provenance

The proof is split between human-checkable analytic reductions and a substantial exact-rational validation code base. The checker implements rational interval arithmetic, rational Taylor or Lagrange enclosures for the required transcendental functions, interval automatic differentiation, fixed-grid outward rounding, and exact comparison of the radii-polynomial inequalities. It does not call a floating transcendental library in a theorem comparison.

The two primary archives are listed in table 2. The hashes are of the files as distributed with this manuscript’s repository snapshot.

Result	Archive	SHA-256 prefix
Profile and spectral endpoint data	combined AU.1/AU.2 certificate	1d5fe53786cc
Fixed-wall radial gap	radial-gap certificate	695310609d07

Table 2: Primary exact-rational proof archives. Full hashes are recorded in the accompanying manifest and in the text below.

The full digests are

1d5fe53786cc280006d7b1092d360556d4d8d8684e5ae3356ce8cd6d084e72a9,
695310609d070f6dba2c678982ffc87472ed7c341813137fb90ad9dd9e1e6096.

The radial certificate additionally consumes the authenticated sharp-tube snapshot with digest

1781a2ff357f3b165d23e290eb403552d77d099a72b9e90920735e6a80b30431.

The union of application sources pinned by the two certificate ledgers and the sharp-tube snapshot is 14,059 physical lines in this repository snapshot, spread across fourteen modules and audit drivers. It is a source-pinned review surface, not a minimized or formally verified trusted kernel: it includes proposal and serialization paths, while the interpreter and standard library remain environmental assumptions. The certificate records a SHA-256 digest for every file in this application surface. Floating-point routines may propose splines and meshes; exact-rational validation modules recompute every proof-relevant enclosure and closing inequality.

Primary reviewer role	Ledger-pinned source group	Physical lines
Candidate generation	qgtoy/skymion_global_bvp_certificate_generator.py; proposal only	1,303
Exact validation and reconstruction	nine modules: rational arithmetic/text, BVP/origin, spectral, radial-gap, and sharp-profile validation	11,024
Closing theorem comparison	three-validator subset: spectral derivatives/ledger and radial gap	subset
Presentation and serialization	decimal summaries and manuscript; no theorem authority or pinned plotting module	0 additional
Authentication, audit, and replay	four experiments/skymion*_audit.py drivers	1,732
Ledger-pinned application surface	fourteen files	14,059

Table 3: Roles in the source-pinned review surface. Roles can overlap: the closing-comparison row identifies a proof-critical subset of the exact validation row and is not counted twice. Source hashes establish provenance, not correctness by themselves.

From the repository root, the focused validation path is

```
python -m pip install -e . 'pytest>=8,<10'
PYTHONPATH=. python experiments/skymion_newton_linearization_audit.py \
  --omega 3/4 --tube-radius 1/250 \
  --tube-trigonometric-terms 12 \
  --tube-rounding-denominator 1000000000000000000 \
  --spectral-trigonometric-terms 24 \
  --spectral-atanh-terms 80 --spectral-pi-terms 80 \
  --origin-kernel-terms 20 \
  --output /tmp/skymion_au2_replay_summary.json \
  --exact-certificate-output /tmp/skymion_au2_replay.json
python paper/validated_skymion_profile/audit_package.py \
  --replay-au2 /tmp/skymion_au2_replay.json
PYTHONPATH=. python experiments/skymion_au3b_sharp_snapshot_audit.py \
  --canonical-au2 \
    experiments/skymion_au2_global_tail_exact_certificate.json \
  --output /tmp/skymion_sharp_snapshot_replay.json
PYTHONPATH=. python experiments/skymion_full_radial_gap_audit.py \
  --output /tmp/skymion_radial_gap_replay.json
sha256sum \
  /tmp/skymion_sharp_snapshot_replay.json \
  /tmp/skymion_radial_gap_replay.json
python -m pytest -q \
  tests/test_validated_skymion_profile.py \
  tests/test_validated_skymion_bvp.py \
  tests/test_validated_skymion_spectral_derivatives.py \
  tests/test_validated_skymion_spectral_ledger.py \
  tests/test_static_patch_skymion_tail.py \
  tests/test_skymion_radial_dynamical_gap.py \
  tests/test_validated_skymion_radial_gap.py \
  tests/test_validated_rational_text.py \
  tests/test_validated_skymion_profile_manuscript.py
```

The two hashes printed by sha256sum must be the sharp-snapshot and radial values displayed above. The combined AU.1/AU.2 archive also records the literal command and a dependency ledger. Its full-file hash is therefore environment-sensitive. A byte-identical clean replay used the archived output paths, Python 3.12.13, NumPy 2.2.6, and no SciPy.

Across supported environments, the audit removes only the top-level provenance fields `command` and `dependency_versions`, then checks the canonical mathematical-payload digest

f1703ab8f2b3f908a1609d03349c60becdb287225374c91c136fb275053d0f85.

All exact inputs, source hashes, parameters, and theorem outputs remain inside that comparison. The source, certificate archives, manifest, and manuscript are distributed at

<https://github.com/bfioca/physics-and-information-theory>.

The immutable external-review snapshot is the Git tag `paper-a-review-v1`. A later journal version should cite its release tag or archival DOI; the file-level hashes above identify the present referee snap-

shot independently of mutable branch names. The exact test filenames and a machine-readable manifest are checked as part of the release audit; no unrelated observer-algebra test is a premise of the paper.

The archive format records both exact fractions and downward/upward decimal summaries. Decimal values printed in the manuscript are presentation layers, not inputs to subsequent proof steps. Every composed certificate checks the identity of its upstream archive before importing a wall slope, inertia, endpoint jet, or Newton radius.

6 Discussion and limitations

The result establishes a nontrivial confined solution with authenticated local uniqueness and several physical diagnostics. It does not establish that the chosen boundary condition is dynamically generated. The imposed Dirichlet condition fixes topological charge, represents external support, enters the radial fluctuation domain, and determines the leading ultraviolet phase and coefficient. The support traction and its dynamics are not derived. The boundary condition is therefore part of the mathematical model rather than a numerical convenience.

This distinction matters for smoothing. A generic finite-stiffness Robin condition is not automatically a nearby realization in the same exact topological sector: imposing both the exact boundary value needed for unit charge and the associated finite-stiffness condition can force vanishing value and derivative, hence the trivial solution by ODE uniqueness. A controlled smooth-confinement limit would require explicit boundary degrees of freedom or a topology-preserving matter-plus-boundary action. Standard Fourier endpoint theory then predicts how the tail changes, but such a family is not constructed here.

The fixed-wall radial gap is similarly specific. It excludes an unstable spherically symmetric profile fluctuation on the declared form domain. It does not control a moving membrane, nonspherical grand-spin channels, an anchor or holding apparatus, or the coupled metric. No claim of full dynamical stability follows.

Finally, the metric is prescribed. We do not solve the Einstein–Skyrme constraints, impose Israel matching conditions, or prove weak gravitational backreaction. The rotor inertia and its finite optical transform make the profile suitable as an input to those questions, but they do not answer them. Turning that transform into a bath factor would require a specified current, interaction, switching protocol, and open-system limit.

The analytic continuation result in corollary 2.2 proves that the solution is not isolated in parameter space, but it supplies no numerical width. The natural next mathematical step is an explicit rational parameter box, with the same certificate architecture and a carefully modeled boundary. The natural physical step is a local matter-plus-support action whose static solution reduces to the validated branch and whose radial, nonspherical, and gravitational perturbations can be controlled together. Those extensions would be new theorems, not consequences silently imported into the present one.

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